Macroeconomics I — MPE Insper

Practice list: Measurement, Solow and Growth Accounting (Listas 1–2)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Measuring output, prices and welfare *(Kurlat chs. 1–2 • 25 points)*

Question 1. (10 points) Objective items. IBGE's national accounts team tests its procedures on a two-good economy (food, good 1; electronics, good 2):

	p_1	q_1	p_2	q_2
2024	2	10	10	5
2025	3	12	7	8

a) (3 points) With 2024 as the base year, the GDP deflator for 2025 (2024 = 100) is:

A. 92.86

B. 131.43

C. 88.46

D. 113.04

b) (3 points) Real GDP growth between 2024 and 2025 computed with Kurlat's chained method (geometric average of the growth rates at initial-year and final-year prices) is:

A. 48.57%

B. 31.43%

C. 41.54%

D. 45.01%

c) (2 points, T or F) If the IPCA, a Laspeyres index with a fixed basket, rises 6% in a year, then the extra income a household needs to reach its original utility level rose by *at least* 6%.

d) (2 points, T or F) Suppose tradable goods obey the law of one price at the market exchange rate and non-tradable services are cheaper in Brazil than in the US. Then converting Brazil's GDP to dollars at the market exchange rate understates it relative to a PPP conversion.

Question 2. (15 points) Discursive. A Penn World Table analyst compares Brazil and the US. Each produces a tradable good T and a non-tradable service N. Prices are in local currency; the market exchange rate is $e = 5$ reais per dollar.

	p_T	q_T	p_N	q_N
Brazil (reais)	10	50	20	100
US (dollars)	2	200	20	150

a) (5 points) Compute Brazil's GDP in dollars (i) at the market exchange rate and (ii) at PPP, using US prices as the common prices. Compute Brazil/US GDP under each measure and the implied PPP exchange rate. Explain the gap.

Solution:

b) (5 points) Now use the Jones–Klenow welfare measure of Kurlat §2.2 with $\sigma = 1$ and equal leisure in both countries, so that a random person's utility is $\mathbb{E}\big[a(\bar{u} + \ln c)\big]$, where $a = 1$ if alive. Normalize $c_{US} = 1$, set $\bar{u} = 5$, survival probabilities $a_{US} = 0.79$, $a_{BR} = 0.76$, and Brazilian consumption per person $\bar{c}_{BR} = 0.30$, *equal for everyone*. Derive λ (the scaling of US consumption that leaves Rawls indifferent), compute it, and split the welfare gap into a consumption part and a life-expectancy part.

Solution:

c) (5 points) Now half of Brazilians consume 0.12 and half 0.48 (same mean 0.30). Recompute λ. By how much does inequality lower it? Would a higher σ make the penalty larger or smaller? Why does Brazil fall further behind in welfare than in consumption?

Solution:

Block 2 — Solow, the Golden Rule and growth accounting *(Kurlat chs. 4–5 • 25 points)*

Question 3. (10 points) Objective items. IPEA calibrates a Solow model *per worker*, without technological progress: $y = k^{\alpha}$, $\alpha = 1/3$, $s = 0.20$, $n = 0.02$, $\delta = 0.08$.

a) (3 points) Steady-state output per worker y^* is:

A. 2.828

B. 1.131

C. 1.414

D. 2.000

b) (3 points) Moving the economy to the Golden Rule saving rate raises steady-state consumption per worker by approximately:

A. 7.6%

B. 0%, since $s = 0.20$ already maximizes c^*

C. 33.3%

D. 16.7%

c) (2 points, T or F) Because this economy is below the Golden Rule, raising s to s_{gr} is a Pareto improvement.

d) (2 points, T or F) If $g_Y = 3\%$, $g_K = 4.5\%$, $g_L = 1.5\%$ and $\alpha = 1/3$, then all growth of output per worker comes from capital deepening.

Question 4. (15 points) Discursive. Brazil is modeled as a Solow economy with labor-augmenting technology: $Y = K^{\alpha}(AL)^{1-\alpha}$, $\alpha = 1/3$, $s = 0.24$, $n = 0.01$, $g = 0.02$ (growth of A), $\delta = 0.05$. Work *per efficiency unit*: $\tilde{k} = K/(AL)$, $\tilde{y} = Y/(AL)$.

a) (5 points) Derive the law of motion of $\tilde{k}$ (continuous time) and compute $\tilde{k}^*$, $\tilde{y}^*$, $\tilde{c}^*$, K/Y and the marginal product of capital in steady state. Give the growth rates of $\tilde{y}$, Y/L and Y on the balanced growth path. Is Brazil above or below the Golden Rule? Prove it with the marginal product, then confirm with $\tilde{c}$.

Solution:

b) (5 points) Floods destroy 30% of the capital stock at t_0 (illustrative). Using the log-linear approximation $\dot{\ln \tilde{y}} \approx -\lambda(\ln \tilde{y} - \ln \tilde{y}^*)$, derive λ, compute the half-life of the gap, the impact effect on Y/L and its growth rate just after the shock. Separately, suppose emigration permanently lowers n to 0.005: compute the change in $\tilde{y}^*$ and the long-run growth of Y/L and of Y.

Solution:

c) (5 points) Development accounting with $y = A\,k^{\alpha}h^{1-\alpha}$. Brazil/US ratios: $y = 0.25$, $k = 0.20$, $h = 0.60$. Compute the TFP ratio and the share of the log income gap due to k, h and TFP. Then suppose a quarter of Brazil's measured capital is security capital (walls, armored cars) that produces nothing. Recompute the TFP ratio on productive capital and explain what the residual measured in the first step was capturing.

Solution: